

UNIVERSITY OF NORTHERN BRITISH COLUMBIA

ECON 712: APPLIED ECONOMETRICS

Does a Carbon Tax Reduce CO₂ Emissions? Evidence from British Columbia

A Replication Paper

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1 Summary and Contribution

1.1 Research Question

The paper by [Pretis \(2022\)](#) examines whether the introduction of a carbon tax in British Columbia (BC) in 2008 led to a reduction in CO₂ emissions. More specifically, it investigates whether carbon pricing can generate measurable reductions in emissions at the aggregate level, rather than only within particular sectors.

The policy motivation for this question arises from the growing use of carbon taxes to address the negative external effects associated with emissions. Carbon pricing is widely viewed as an efficient policy tool for reducing emissions ([Carattini et al. 2019](#)). However, there is limited empirical evidence on whether such policies lead to reductions in total emissions, as opposed to affecting only specific sectors. Since climate change is driven by cumulative emissions, it is important to understand the overall impact of these policies.

1.2 Contribution to the Literature

The paper contributes to the literature in two main ways. It places emphasis on aggregate emissions rather than sector specific outcomes. Much of the existing literature shows that carbon pricing reduces emissions in particular sectors, especially transportation ([Andersson 2019](#), [Davis & Kilian 2011](#)). However, these findings do not necessarily imply that total emissions decline, since reductions in one sector may be offset by increases in others. By focusing on aggregate emissions, the paper addresses a gap that is directly relevant for evaluating climate policy.

The paper also contributes methodologically by combining several empirical approaches. While the main analysis is based on a difference in differences framework, it is complemented by synthetic control methods and a break detection approach based on indicator saturation ([Castle et al. 2015](#)). This allows the analysis not only to estimate the effect of a known policy, but also to identify other structural changes in emissions that may not be directly linked to carbon pricing.

An additional contribution lies in how the paper addresses inference in the presence of a single treated unit. In such settings, conventional methods may be unreliable, so the au-

thor uses randomization inference following (Cameron et al. 2008). This strengthens the credibility of the results by providing an alternative way to assess statistical significance.

Despite these strengths, the approach has some limitations. Identification relies on the parallel trends assumption, which may not hold if British Columbia differs from other provinces in ways that affect emissions trends. The presence of only one treated unit also limits statistical power and makes it more difficult to obtain precise estimates.

1.3 Main Findings

The main finding of the paper is that the carbon tax in British Columbia did not lead to a statistically significant reduction in total CO₂ emissions. At the same time, there is clear evidence of a decline in transportation emissions, consistent with earlier studies on carbon pricing (Andersson 2019, Metcalf & Stock 2020).

This pattern reflects differences across emission sources. While transportation shows a noticeable response, other sectors such as stationary energy and industrial processes exhibit little change, and in some cases even slight increases. As a result, the overall impact at the aggregate level remains small and imprecisely estimated.

The results from this replication are consistent with those in the original paper. Using an extended dataset covering 1990 to 2023, the estimated effect of the carbon tax on total emissions is negative but not statistically significant. In contrast, the transportation sector shows a statistically significant reduction in emissions, both in the short run and in the long run after accounting for dynamic adjustment. Other sectors show negative but statistically insignificant effects.

The paper interprets these results as causal based on the difference in differences framework. This relies on the assumption that, in the absence of the carbon tax, emissions in BC would have followed the same trend as in the control provinces. Although this assumption cannot be tested directly, the paper provides supporting evidence through graphical analysis and robustness checks. However, the results should still be interpreted with caution, given the non random assignment of the policy and the limited statistical power of the analysis.

2 Identification Strategy

2.1 The Identification Problem

The main identification challenge in this paper arises from the fact that the carbon tax in British Columbia was not randomly assigned. British Columbia may differ from other provinces in ways that also influence emissions, including differences in industrial structure, environmental preferences, and policy environment. As a result, a simple comparison of emissions before and after 2008, or between BC and other provinces, would not provide a causal estimate of the policy effect.

Such comparisons are likely to be affected by several sources of bias. Emissions may follow different underlying trends across provinces, so changes observed after the policy could reflect pre-existing dynamics rather than the impact of the tax. For instance, if BC was already experiencing slower emissions growth before 2008, attributing post-policy changes to the carbon tax would overstate its effect. In addition, macroeconomic shocks, such as the 2008 - 2009 financial crisis, may affect provinces differently depending on their economic structure. Since emissions are closely linked to economic activity, failing to account for these shocks could bias the estimates. There are also structural differences across provinces, including variation in energy use and industrial composition, which can lead to systematic differences in emissions ([Pretis 2022](#)).

These considerations highlight the need for an empirical strategy that accounts for both time invariant differences across provinces and common shocks over time in order to isolate the causal effect of the carbon tax ([Angrist & Pischke 2009](#)).

2.2 The Identification Strategy

To address these challenges, the paper uses a difference in differences framework ([Angrist & Pischke 2009](#)). The idea is to compare the change in emissions in British Columbia after the introduction of the carbon tax in 2008 to the change in emissions in other provinces over the same period.

British Columbia is treated as the treated unit, while the remaining provinces serve as the control group. The treatment variable is defined as an interaction between an indicator

for BC and an indicator for the post 2008 period. This captures the exposure of BC to the carbon tax after its introduction.

The key identifying assumption in this framework is the parallel trends assumption. This means that, in the absence of the carbon tax, emissions in British Columbia would have followed a similar trend to those in the other provinces. Under this assumption, any difference in emissions between BC and the control group after 2008 can be attributed to the policy. The paper provides some support for this assumption by examining pre-treatment trends and showing that there are no clear differences before the tax was introduced. In addition, the specification includes province and year fixed effects. Province fixed effects (μ_i)² account for persistent differences across provinces, such as differences in industrial structure or geography, while year fixed effects (λ_t)² capture shocks that affect all provinces in a given year, including macroeconomic conditions and changes in energy markets (Pretis 2022). Because there is only one treated unit, standard inference may be unreliable. To address this, the paper uses randomisation inference, which compares the estimated treatment effect to placebo estimates obtained by assigning treatment to other provinces (Cameron et al. 2008, Pretis 2022).

2.3 Empirical Specification

To estimate the effect of the British Columbia carbon tax on CO₂ emissions, the paper uses a dynamic difference in differences framework following (Pretis 2022). The analysis is based on an annual panel of Canadian provinces covering the period 1990 - 2023. British Columbia is treated following the introduction of the carbon tax in 2008, while the remaining provinces serve as the control group. The treatment variable is defined as

$$Tax_{i,t} = Treat_i \times Post_t \quad (1)$$

where $Treat_i$ equals one for British Columbia and zero otherwise, and $Post_t$ equals one for years beginning in 2008. This interaction captures exposure to the policy after its introduction.

The empirical model² relates the logarithm of emissions to the treatment indicator, lagged emissions, and a set of controls, while accounting for both province and year fixed effects.

The estimating equation is given by

$$\log(Emiss)_{i,t} = \lambda_t + \mu_i + \beta Tax_{i,t} + \alpha \log(Emiss)_{i,t-1} + \gamma_1 \log(Pop)_{i,t} + \gamma_2 \log(GDP)_{i,t} + \varepsilon_{i,t}. \quad (2)$$

Here, $Emiss_{i,t}$ denotes CO₂ emissions in province i and year t . Province fixed effects (μ_i) control for time invariant differences across provinces, such as structural differences in economic activity, geography, and industrial composition. Year fixed effects (λ_t) capture shocks common to all provinces in a given year, including macroeconomic conditions and changes in energy markets.

Population and real GDP are included to account for differences in scale and economic activity across provinces over time. The inclusion of lagged emissions allows for persistence in emission dynamics, reflecting the gradual adjustment of production processes, infrastructure, and energy use.

In this dynamic setting, the coefficient β measures the short run effect of the carbon tax. Because emissions adjust over time through the autoregressive structure, the long run or equilibrium effect is given by

$$\beta^{eqbm} = \frac{\beta}{1 - \alpha}. \quad (3)$$

This expression captures the total adjustment in emissions once the dynamic response has fully materialized. Since the equilibrium effect is a nonlinear function of estimated parameters, its standard error is obtained using the delta method.

Identification relies on the difference in differences assumption that, conditional on fixed effects and included controls, emissions in British Columbia would have followed the same trend as those in the control provinces in the absence of the carbon tax. While this assumption cannot be tested directly, the data do not show clear evidence of diverging trends prior to the policy. To allow for differences across emission sources, the model is estimated not only for total emissions but also separately for major sectors defined under the IPCC classification. This makes it possible to assess whether the impact of the carbon tax varies across sectors with different energy use patterns and adjustment possibilities. The extended sample period used in this replication increases the number of

post treatment observations and allows the dynamic response of emissions to be evaluated over a longer horizon.

While the specification includes key controls for economic activity and population, some relevant factors may still be omitted. In particular, changes in energy prices and the presence of other environmental policies could influence emissions and may vary across provinces over time. If these factors are correlated with both the carbon tax and emissions, they could bias the estimated effect. Although fixed effects account for time invariant differences and common shocks, they may not fully capture time varying province specific influences.

2.4 Critical Assessment

The identification strategy is well grounded and follows standard approaches in applied econometrics. The use of a difference in differences framework, together with fixed effects and relevant controls, provides a sensible way to isolate the effect of the carbon tax. At the same time, there are some limitations that are difficult to ignore. The analysis relies on a single treated unit, which makes it harder to obtain precise estimates and raises concerns about the reliability of inference, even when using clustered or randomisation based methods ([Cameron et al. 2008](#)). In addition, the parallel trends assumption cannot be verified directly. Although the paper presents graphical evidence and a range of robustness checks, it is still possible that British Columbia differs from other provinces in ways that affect emissions independently of the policy.

There are also some modelling considerations. The inclusion of a lagged dependent variable is useful for capturing persistence in emissions, but it may introduce bias in a fixed effects setting, especially in panels with a limited time dimension ([Nickell 1981](#)). While this concern is less severe in this case given the relatively long time span of the data, it cannot be ruled out completely. Moreover, other factors such as changes in industrial activity, energy prices, or concurrent policies may have influenced emissions during the period under study, and these effects may not be fully captured by the controls included in the model. For these reasons, the results are informative and broadly consistent with a causal interpretation, but they should still be viewed with some caution.

3 Replication

3.1 Data

The analysis is based on annual panel data on provincial CO₂ emissions, economic activity, and population for Canadian provinces. Emissions data are obtained from the official Greenhouse Gas Inventory Data Portal of Environment and Climate Change Canada, specifically from the dataset *EN_GHG_IPCC_Can_Prov_Terr.csv*. The dataset is available at:

<https://data-donnees.az.ec.gc.ca/data/substances/monitor/canada-s-official-greenhouse-gas-inventory/A-IPCC-Sector/>

This dataset reports emissions by province, year, gas type, and IPCC source category. Following Pretis (2022), the data are restricted to CO₂ emissions and aggregated according to the IPCC sector classifications, including total emissions, stationary energy, transportation, fugitive emissions, industrial processes, agriculture, and waste. Data on provincial GDP and population are taken from Statistics Canada. GDP is measured in constant 2017 dollars and is included to capture macroeconomic conditions, while population accounts for differences in the scale of economic activity across provinces. These variables are included in line with the original study, where GDP is used in part to capture the effects of the 2008 - 2009 financial crisis on emissions Pretis (2022).

The original paper uses data covering the period 1990 - 2016. In this replication, the dataset is extended to include observations up to 2023. The sample includes ten provinces, excluding the northern territories due to data limitations. This produces a balanced panel with 340 province-year observations, compared to 270 observations in the original study.

Constructing the dataset required filtering emissions to retain CO₂ only, aggregating observations to the province-year-sector level, and merging emissions data with GDP and population using consistent identifiers. Ensuring consistency across datasets required careful alignment of units and definitions. The resulting dataset closely follows the structure used in Pretis (2022), with the main difference being the longer time horizon.

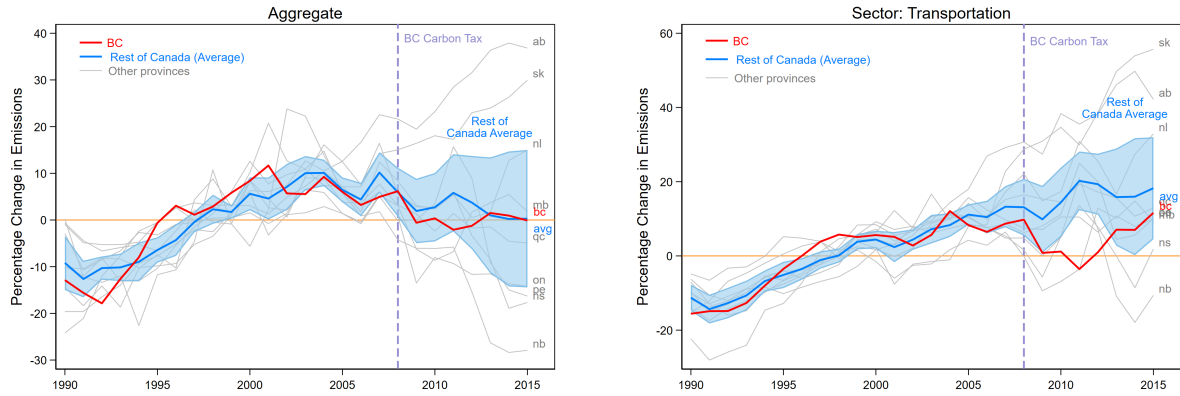


Figure I: Percentage change in province-level aggregate CO₂ emissions relative to the 1990–2007 average for aggregate emissions (left) and transportation emissions (right, the single largest emitting sector in the taxed region). The blue solid line corresponds to the average across included provinces other than British Columbia (BC), which is shown in red. The carbon tax is indicated by a purple dashed line. The blue shading represents the 95% confidence interval of the “rest of Canada” average in each period. Canadian provinces are abbreviated as follows: British Columbia (bc), Alberta (ab), Ontario (on), Newfoundland and Labrador (nl), Nova Scotia (ns), New Brunswick (nb), Saskatchewan (sk), Quebec (qc), Prince Edward Island (pe), and Manitoba (mb).

Table 1: Province-level summary statistics for taxed region (BC) and control provinces 1990 - 2023

	CO ₂ emissions							Controls	
	Total	Stat. energy	Transp.	Fugit	Indust	Agric	Waste	Pop	GDP
<i>BC (taxed)</i>									
Mean	47.40	44.97	22.88	2.07	2.41	0.03	0.00	4.35	221.14
SD	3.45	3.30	2.57	0.80	0.40	0.01	0.00	0.57	58.65
Max	52.03	49.38	27.62	3.17	3.32	0.05	0.00	5.52	343.02
Min	38.28	36.24	18.17	0.75	1.61	0.01	0.00	3.29	137.55
<i>Control prov</i>									
Mean	54.60	50.10	16.78	1.45	4.26	0.21	0.02	3.18	168.48
SD	63.60	57.98	17.55	3.08	6.54	0.25	0.04	4.08	220.97
Max	219.22	207.62	61.24	12.73	22.59	1.22	0.12	15.63	947.16
Min	1.13	1.12	0.58	0.00	0.00	0.00	0.00	0.13	3.66

Emissions measured in Mt CO₂; population in millions; GDP, billions in constant 2017 dollars. Control provinces here include: Alberta (ab), Ontario (on), Newfoundland and Labrador (nl), Nova Scotia (ns), New Brunswick (nb), Saskatchewan (sk), Quebec (qc), Prince Edward Island (pe), Manitoba (mb). Estimation results with different controls (omitting those that have carbon pricing or trading schemes) are reported in Supplementary Material.

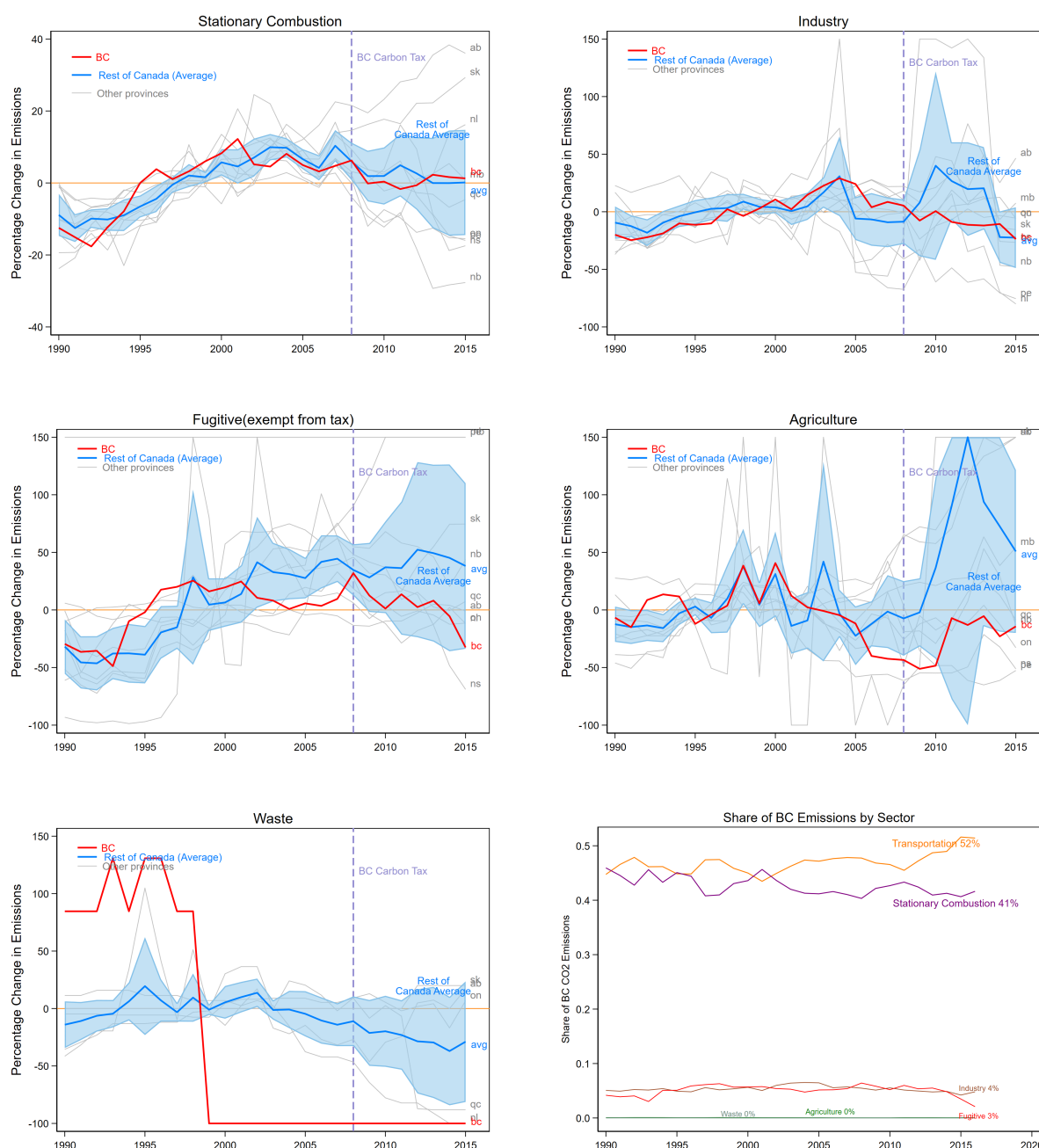


Figure II: Percentage change in province-level, sector-specific CO₂ emissions relative to the 1990–2007 average. Sector definitions correspond to IPCC emission categories. The blue solid line shows the average across included provinces other than British Columbia, which is shown in red. The blue shading represents the 95% confidence interval of the “rest of Canada” average in each period. The lower-right panel shows the share of emissions in the taxed province (British Columbia) by sector over time.

3.2 Descriptive Statistics

The data show clear differences between British Columbia and the control group. Average total emissions in BC are lower than in the control provinces, which reflects differences in economic structure and energy use (see Table 1). At the same time, transportation emissions represent a relatively large share of emissions in BC, suggesting that this sector plays an important role in the province's emissions profile.

The control provinces exhibit substantial variation in emissions, as reflected in the large standard deviations. This is consistent with the diversity of provincial economies, particularly the presence of resource intensive provinces such as Alberta and Saskatchewan. These differences highlight the importance of controlling for both province specific characteristics and time varying factors in the empirical analysis.

These patterns are broadly consistent with those reported in [Pretis \(2022\)](#), which provides reassurance that the data have been constructed correctly and are comparable to those used in the original study.

3.3 Main Results

The main regression results from the difference in differences estimation are presented in Table 2.

The estimated effect of the carbon tax on total CO₂ emissions is negative but not statistically significant. In the replication, the short run coefficient is approximately -0.007 with a standard error of about 0.006, corresponding to a reduction of about 0.7 percent. When the dynamic structure of the model is taken into account, the implied long run effect is approximately -6.3 percent, although this estimate remains statistically insignificant.

This pattern closely matches the findings in [Pretis \(2022\)](#), where the effect on aggregate emissions is also imprecisely estimated. The lack of statistical significance suggests that the carbon tax did not produce a sufficiently large change in total emissions to be detected at the aggregate level.

The sectoral results show a different pattern. The transportation sector exhibits a statistically significant reduction in emissions, with an estimated short run effect of approx-

imately -2.2 percent and a larger long run effect once dynamic adjustment is taken into account. This finding is consistent with [Pretis \(2022\)](#) and aligns with previous evidence that carbon pricing tends to have stronger effects in transportation ([Andersson 2019](#)) and ([Metcalf & Stock 2020](#)).

Table 2: The effect of the carbon tax on CO₂ emissions (difference-in-differences estimation, 1990-2023)

	Dependent variable: $\log(Emissions)_{it}$						
	Total	Stat. Energy	Transportation	Fugit. (excmp)	Industry	Agriculture	Waste
Carbon Tax _{it}	-0.0071 (0.0056)	-0.0040 (0.0055)	-0.0223*** (0.0066)	-0.0673 (0.0652)	-0.0882 (0.0572)	-0.1590 (0.1259)	—
$\log(Emiss)_{i,t-1}$	0.887*** (0.0386)	0.880*** (0.0392)	0.844*** (0.0487)	0.962*** (0.0194)	0.714*** (0.0741)	0.687*** (0.1182)	0.921*** (0.0455)
$\log(Pop)_{it}$	-0.007 (0.0417)	-0.008 (0.0489)	-0.095** (0.0335)	0.612** (0.2558)	0.746** (0.2800)	-0.785 (0.9035)	0.458 (0.3304)
$\log(GDP)_{it}$	0.226** (0.0764)	0.229** (0.0792)	0.314*** (0.0898)	0.230 (0.7590)	0.121 (0.3643)	1.688 (1.2321)	-0.243 (0.3009)
Eqbm. Tax Coef	-0.0630 (0.0459)	-0.0331 (0.0438)	-0.1429*** (0.0321)	-1.7726 (2.2388)	-0.3087* (0.1728)	-0.5082** (0.2396)	—
Pct. Impact	-0.7% (0.6%)	-0.4% (0.6%)	-2.2%*** (0.7%)	-6.5% (6.1%)	-8.4% (5.2%)	-14.7% (10.7%)	—
Pct. Impact Se							
Pct. Eqbm. Impact	-6.1% (4.3%)	-3.3% (4.2%)	-13.3%*** (2.8%)	-83.0% (38.0%)	-26.6%* (12.7%)	-39.8%** (14.4%)	—
Pct. Eqbm. Impact Se							
Observations	330	330	330	297	330	323	170
Clusters	10	10	10	9	10	10	7
R^2 (within)	0.860	0.852	0.840	0.846	0.601	0.614	0.831
F Statistic	562.46***	741.75***	253.45***	1724.97***	118.94***	349.28***	9362.85***

Province and year fixed effects included. Standard errors clustered at the province level (in parentheses). Equilibrium impact equals $\beta/(1 - \rho)$. Percent impacts are computed as $100(e^\beta - 1)$. Percent impact standard errors are obtained using the delta method transformation. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The number of observations and clusters varies across specifications due to the dynamic nature of the model and differences in data availability across sectors. The inclusion of lagged emissions reduces the sample size by one year for each province. In addition, the use of logarithmic transformations leads to the exclusion of observations with zero

or missing emissions. In some sectoral regressions, this results in entire provinces being dropped, which reduces the number of clusters.

Other sectors, including stationary energy, industrial processes, and fugitive emissions, show negative point estimates but are not statistically significant. This suggests that the impact of the carbon tax varies across sectors and that reductions in transportation emissions are not sufficient to generate a detectable decline in total emissions.

Additional specifications produce similar results. Restricting the control group by excluding Alberta and Quebec does not materially change the estimates, although precision is reduced (see Appendix Table A1). Using different inference methods, including conventional standard errors (see Appendix Table A2) and bootstrap procedures (see Appendix Table A3), leads to similar point estimates, with statistical significance remaining limited outside the transportation sector. Alberta and Quebec are excluded from the control group due to their distinct economic and energy structures. Alberta's emissions are heavily influenced by oil and gas production, while Quebec relies largely on hydroelectric power and has a much lower emissions intensity.

The randomization inference results in Figure III reinforce this pattern. The placebo distribution is constructed by repeatedly assigning the treatment to non-treated provinces and re-estimating the model, thereby capturing the range of effects that could arise by chance. The estimated effect for transportation emissions lies in the tails of this distribution, indicating that it is unlikely to have occurred due to random variation and providing strong evidence against the null hypothesis of no effect. In contrast, the estimate for aggregate emissions falls well within the range of placebo estimates, suggesting that it is not statistically distinguishable from zero. This is consistent with the regression results, which show a significant effect for transportation emissions but not for aggregate emissions.

Additional robustness checks are reported in Appendix A. The results remain broadly consistent across specifications. The replication results are closely aligned with those reported in (Pretis 2022). Differences in magnitude are small and can be explained by the extended sample period and updates to the underlying data, without altering the main conclusions.

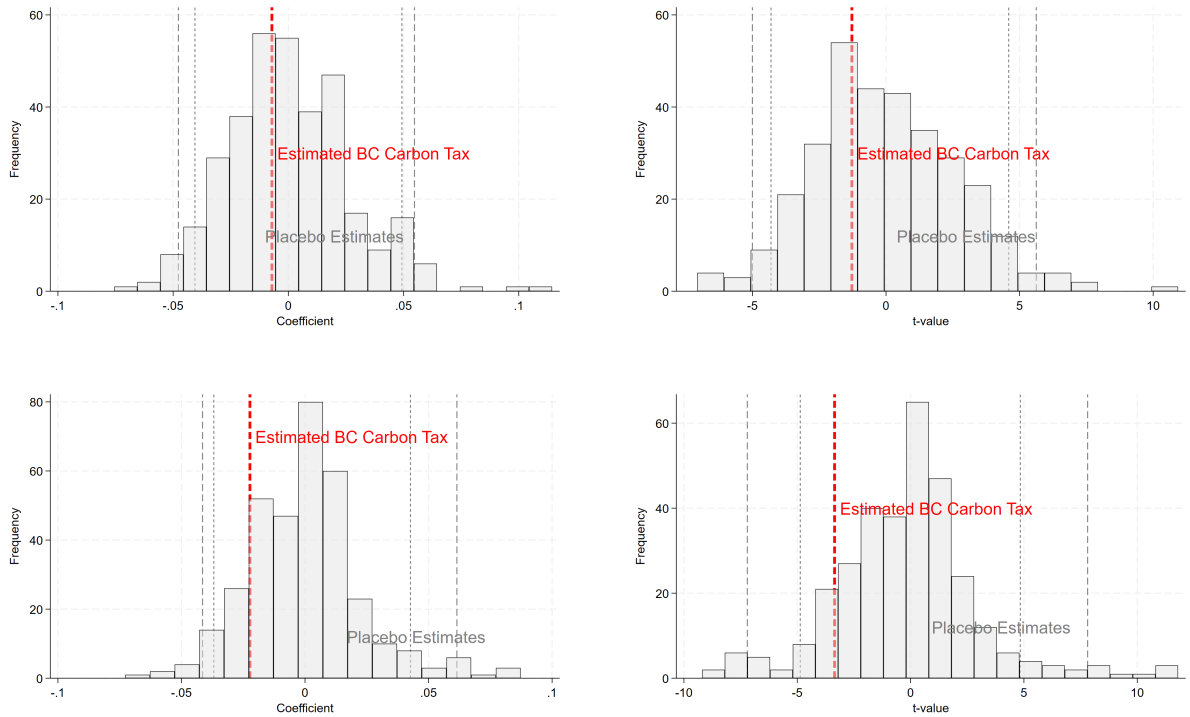


Figure III: Randomization inference results. The top panels show aggregate CO₂ emissions, while the bottom panels show transportation CO₂ emissions. In each case, the histograms display the distribution of placebo treatment effects, and the vertical red line indicates the actual British Columbia carbon tax estimate.

4 Critical Discussion and Proposed Extensions

4.1 Assessment of Empirical Choices

The paper makes several empirical choices that are generally well motivated, although some raise important considerations. The use of CO₂ emissions as the outcome variable is appropriate given the policy question, since climate change is driven by total emissions. At the same time, reporting results at the sectoral level is particularly informative, as it reveals heterogeneity across emission sources, with stronger effects observed in transportation. The restriction to the ten Canadian provinces is reasonable given data limitations for the northern territories, but the small number of cross sectional units creates challenges for inference, especially with a single treated unit. The inclusion of GDP and population as controls is well justified, as these variables capture macroeconomic conditions and differences in scale across provinces. However, other relevant factors, such as energy prices or technological change, are not explicitly accounted for.

Clustering standard errors at the province level follows standard practice, although with

only ten clusters inference may be unreliable (Cameron et al. 2008). The use of randomisation inference is therefore a strength, as it provides a more appropriate way to assess statistical significance in this setting. The log linear specification allows for a clear percentage interpretation of coefficients, and the inclusion of lagged emissions captures persistence in emission dynamics. This comes at the cost of potential Nickell bias in fixed effects estimation Nickell (1981), although this concern is reduced by the relatively long time dimension of the data.

4.2 Proposed Improvements

One useful extension would be to allow for heterogeneous treatment effects. The impact of the carbon tax may vary across provinces depending on industrial structure or energy intensity. This could be examined by interacting the treatment variable with measures of sectoral composition or energy use, providing a more detailed understanding of where the policy is most effective.

Another improvement would be to rely more heavily on alternative identification strategies suited to a single treated unit. Methods such as synthetic control can construct a more credible counterfactual for British Columbia by matching pre treatment trends more closely. While this approach is used in the paper, further development and comparison across specifications could strengthen the results.

A further extension would be to use more disaggregated data, such as firm or plant level emissions. Province level data may mask important within province variation, and more detailed data could provide additional insight into the mechanisms through which the carbon tax affects emissions.

4.3 Overall Assessment

The paper provides a careful empirical analysis of the British Columbia carbon tax and offers useful evidence on its effects. The finding that transportation emissions decline while aggregate emissions do not is consistent with both the original study and the replication results. This suggests that the policy had uneven effects across sectors and may not have been strong enough to produce a measurable reduction in total emissions.

While the identification strategy is reasonable, it relies on assumptions that are difficult to verify, particularly given the presence of a single treated unit. As a result, the findings are informative but should be interpreted with some caution. Even so, the paper makes an important contribution by showing that sector specific responses do not necessarily translate into aggregate emission reductions.

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A Additional Results

A.1 Restricted Control Group

Table A1: The effect of the carbon tax on CO₂ emissions excluding Alberta and Quebec from the control group

	<i>Dependent variable: $\log(Emissions)_{it}$</i>						
	Total	Stat. Energy	Transportation	Fugit. (excmp)	Industry	Agriculture	Waste
Carbon Tax _{it}	-0.0054 (0.0124)	-0.0006 (0.0125)	-0.0181* (0.0081)	-0.1265 (0.1399)	-0.0920 (0.0916)	-0.1258 (0.1279)	—
$\log(Emiss)_{i,t-1}$	0.891*** (0.0506)	0.883*** (0.0527)	0.865*** (0.0469)	0.960*** (0.0179)	0.712*** (0.0761)	0.677*** (0.1376)	1.120*** (0.0613)
$\log(Pop)_{it}$	-0.019 (0.0886)	-0.029 (0.0941)	-0.080 (0.0470)	0.729 (0.6020)	0.868*** (0.2584)	-1.168 (1.1361)	-0.028 (0.1298)
$\log(GDP)_{it}$	0.239** (0.0959)	0.242** (0.0992)	0.273** (0.0842)	0.571 (0.9886)	0.060 (0.4113)	1.783 (1.4062)	-0.101 (0.1694)
Eqbm. Tax Coef	-0.0494 (0.1238)	-0.0052 (0.1086)	-0.1338* (0.0781)	-3.1321 (4.6450)	-0.3198 (0.2998)	-0.3896 (0.2911)	—
Pct. Impact	-0.5% (1.2%)	-0.1% (1.2%)	-1.8%* (0.8%)	-11.9% (12.3%)	-8.8% (8.4%)	-11.8% (11.2%)	—
Pct. Impact Se	-4.8% (11.8%)	-0.5% (10.8%)	-12.5%* (6.8%)	-95.6% (20.6%)	-27.4% (22.0%)	-32.3% (19.4%)	—
Observations	264	264	264	231	264	257	104
Clusters	8	8	8	7	8	8	5
R^2 (within)	0.829	0.822	0.819	0.847	0.561	0.604	0.907
F Statistic	473.32***	529.29***	259.62***	2525.05***	58.96***	453.07***	469.17***

Province and year fixed effects included. Standard errors clustered at the province level are reported in parentheses. Equilibrium impact equals $\beta/(1 - \rho)$. Percent impacts are computed as $100(e^\beta - 1)$.

Percent impact standard errors are obtained using the delta method transformation. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.2 Conventional Standard Errors

Table A2: The effect of the carbon tax on CO₂ emissions using conventional standard errors

	<i>Dependent variable: $\log(Emissions)_{it}$</i>						
	Total	Stat. Energy	Transportation	Fugit. (excmp)	Industry	Agriculture	Waste
Carbon Tax _{it}	-0.0071 (0.0170)	-0.0040 (0.0175)	-0.0223 (0.0155)	-0.0673 (0.2010)	-0.0882 (0.0883)	-0.1590 (0.1087)	—
$\log(Emiss)_{i,t-1}$	0.887*** (0.0264)	0.880*** (0.0270)	0.844*** (0.0294)	0.962*** (0.0266)	0.714*** (0.0409)	0.687*** (0.0391)	0.921*** (0.0414)
$\log(Pop)_{it}$	-0.007 (0.0587)	-0.008 (0.0603)	-0.095* (0.0520)	0.612 (0.7116)	0.746** (0.3298)	-0.785** (0.3949)	0.458 (0.3920)
$\log(GDP)_{it}$	0.226*** (0.0713)	0.229*** (0.0737)	0.314*** (0.0650)	0.230 (0.8481)	0.121 (0.3598)	1.688*** (0.4977)	-0.243 (0.4794)
Eqbm. Tax Coef	-0.0630 (0.1484)	-0.0331 (0.1444)	-0.1429 (0.0955)	-1.7726 (5.2978)	-0.3087 (0.3115)	-0.5082 (0.3382)	—
Eqbm. Se							
Pct. Impact	-0.7% (1.7%)	-0.4% (1.7%)	-2.2% (1.5%)	-6.5% (18.8%)	-8.4% (8.1%)	-14.7% (9.3%)	—
Pct. Impact Se							
Pct. Eqbm. Impact	-6.1% (13.9%)	-3.3% (14.0%)	-13.3% (8.3%)	-83.0% (90.0%)	-26.6% (22.9%)	-39.8% (20.3%)	—
Pct. Eqbm. Impact Se							
Observations	330	330	330	297	330	323	170
Clusters	10	10	10	9	10	10	7
R^2 (within)	0.860	0.852	0.840	0.846	0.601	0.614	0.831
F Statistic	435.65***	409.08***	372.15***	347.28***	106.87***	110.01***	209.65***

Province and year fixed effects included. Conventional standard errors are reported in parentheses.

Equilibrium impact equals $\beta/(1 - \rho)$. Percent impacts are computed as $100(e^\beta - 1)$. Percent impact

standard errors are obtained using the delta method transformation. Significance: *** $p < 0.01$, **

$p < 0.05$, * $p < 0.10$.

A.3 Bootstrap Standard Errors

Table A3: Estimated Immediate Level Impact of Carbon Tax (Dif-in-Dif) Cluster Robust Bootstrap Standard Errors (95 percent Confidence Interval)

	Coefficient	Bootstrap SE	p-value	95% CI
Total	-0.0071	(0.0092)	0.437	[-0.0252, 0.0109]
Transportation	-0.0223**	(0.0095)	0.019	[-0.0409, -0.0037]
Fugitive (exempt)	-0.0673	(0.0847)	0.426	[-0.2333, 0.0986]
Stat. Energy	-0.0040	(0.0091)	0.662	[-0.0219, 0.0139]
Industry	-0.0882	(0.0829)	0.287	[-0.2506, 0.0742]
Agriculture	-0.1590	(0.1522)	0.296	[-0.4574, 0.1393]
Waste	—	—	—	—

Notes: Bootstrap standard errors based on 500 replications clustered at the province level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.